

- 8 (a)** Any 1 property of aluminium: B1
1. Strong to withstand stress especially during launch
 2. Tough so will not crack easily
 3. Help to conduct charge from solar wind to sensors
- (b) (i)** Period of earth around sun = $1 \text{ y} = 3.2 \times 10^7 \text{ s}$ M1
- Orbital radius of SOHO = $(150-1.5) \times 10^9 = 1.485 \times 10^{11} \text{ m}$
- Acceleration $a = r\omega^2 = 1.485 \times 10^{11} \left(\frac{2\pi}{365 \times 24 \times 3600} \right)^2$ A1
- $= 5.7 \times 10^{-3} \text{ ms}^{-2}$
- Comment: Many students wrongly stated period of Earth is 1 day or 24 h.
- (ii)** Gravitational force provides the centripetal force. M1
 By Newton's 2nd Law, $F = ma$
 $GMm/r^2 = mr\omega^2$
 $GM/r^3 = (2\pi/T)^2$ M1
 $T^2 = 4\pi^2 r^3 / GM$
- (iii)** Net F_{grav} on SOHO = force due to Sun – force due to Earth M1
 $= Gm(M_s/r_s^2 - M_E/r_E^2)$
 $= 6.67 \times 10^{-11} \times 1900 [(2.0 \times 10^{30} / (1.485 \times 10^{11})^2) - 6.0 \times 10^{24} / (1.5 \times 10^9)^2]$ M1
 $= 11 \text{ N}$
- From **(b)(ii)** Centripetal force = mass x acceleration M1
 $= 1900 \times 5.7 \times 10^{-3}$
 $= 11 \text{ N}$ M1
- Net gravitational force provides the centripetal force as the Earth required to enable SOHO to orbit in 1 year. A1
- (Alternatively, students can show that the period of rotation of SOHO around the Sun is equal to 1 year.)
- (c) (i)** Correct calculation of $1/\lambda_{\text{max}} = 0.0917 \times 10^7 \text{ m}^{-1}$ at $T = 3000 \text{ K}$ A1
- (ii)** Correct plot and best fit line drawn. B1
- (iii)** $k = 1/\text{gradient}$ C1
 $= (5800-2700) / (0.200 \times 10^7 - 0.080 \times 10^7)$
 $= 3100 / (0.120 \times 10^7)$
 $= 2.6 \times 10^{-3} \text{ m K } (\pm 10\%)$ A1
- (d)** soft X-rays range from $(0.10 + 10) \text{ nm}$, average $\lambda = 5.1 \times 10^{-9} \text{ m}$ C1
- Energy of photon = ke of particle

$$hc/\lambda = 3/2 k T$$

$$6.63 \times 10^{-34} \times 3.0 \times 10^8 / 5.1 \times 10^{-9} = 3/2 \times 1.38 \times 10^{-23} T$$

$$T = 1.9 \times 10^6 K$$

C1

A1

(Allow range of values of T from 9.6×10^5 to 9.6×10^7 K)

- (e) (i) Must consider the worse scenario i.e. 3200 kms^{-1}

C1

$$\text{Shortest time } t = \text{distance/fastest speed} = (150 \times 10^9) / (3200 \times 10^3)$$

$$= 46900 \text{ s}$$

$$= 13.0 \text{ h}$$

A1

- (ii) CMEs are charged particles moving at high speeds. So it carries a strong magnetic field with it.

B1

As it travels towards Earth, the changing magnetic field/flux produces induced currents in any closed circuits and therefore make circuit breakers trip.

B1

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MARGIN

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